

HTML, CSS & UI Design Standards

Comprehensive Master Guide: Layouts, Color Themes, Typography Specs & Resources

1. Core HTML & CSS Basics

- **Semantic HTML Structure:** Utilize native semantic layout elements (`<header>` , `<main>` , `<section>` , `<nav>` , `<aside>` , `<footer>`) rather than non-semantic `<div>` wrappers to ensure high accessibility and predictable document architecture.
- **The CSS Box Model:** Understand the structure of every rendered element: **Content** → **Padding** → **Border** → **Margin**. Enforce border-box sizing globally via `box-sizing: border-box` so explicit `width` and `height` properties include padding and borders without overflowing bounds.
- **Modern Layout Engines:** Combine **CSS Grid** for macro-level layouts (dashboard panels, kanban column systems, grid cards) with **CSS Flexbox** for micro-level alignments (navbar content, icon-text alignment, button groups).

2. Color Palettes & Theme Specifications

Light Theme Specification

ELEMENT	HEX CODE	PURPOSE
Background	#F8F9FA	Off-white to reduce glare.
Surface	#FFFFFF	For cards, modals, and raised elements.
Primary Text	#212529	Deep charcoal for high legibility.
Secondary Text	#6C757D	For captions and less important info.
Accent / Brand	#007BFF	Primary interactive elements & active states.

Dark Theme Specification

ELEMENT	HEX CODE	PURPOSE
Background	#121212	Deep grey (Material standard) to allow for shadows.
Surface	#1E1E1E	Slightly lighter to create depth.
Primary Text	#E9ECEF	Soft white to prevent "bleeding" or "halo" effects.
Secondary Text	#ADB5BD	Muted grey for metadata.
Accent / Brand	#375A7F	Desaturated blue (bright colors can be harsh in dark mode).

3. Typography Hierarchy & Rules

STYLE	SIZE (PX)	WEIGHT	USE CASE
Heading 1	32px – 40px	Bold (700)	Page titles
Heading 2	24px – 28px	Semi-Bold (600)	Section headers
Body Large	18px	Regular (400)	Introduction text
Body Main	16px	Regular (400)	Standard reading text
Small / Caption	12px – 14px	Regular (400)	Labels, hints, and footer text

4. Implementation Guidelines

- **Design Tokens & CSS Variables:** Define system values inside the `:root` selector (e.g., `--color-bg-primary`, `--spacing-md`, `--radius-lg`). Toggling light/dark themes will require updating root variables rather than duplicate component classes.
- **Interactive Affordances:** Always provide transition feedback on hover, focus, and active states (e.g., `transition: background-color 0.2s ease, transform 0.1s ease`).
- **Depth & Elevation (Box Shadow):** Indicate interactive layers using shadow scales:
 - **Cards/Containers:** `box-shadow: 0 1px 2px 0 rgba(0, 0, 0, 0.05);`
 - **Modals/Dragging Items:** `box-shadow: 0 20px 25px -5px rgba(0, 0, 0, 0.1);`

5. Design Resources & Reference Links

Color Hunt

<https://share.google/iWuWqMaxgpu8blF9Q>

Flat UI Colors

<https://flatuicolors.com/?hl=en-US>

Google Fonts

<https://fonts.google.com/?hl=en-US>

Dribbble - Light & Dark Theme

<https://dribbble.com/tags/light-dark-theme?hl=en-US>

Adobe Color

<https://color.adobe.com/?hl=en-US>

Typewolf

<https://www.typewolf.com/?hl=en-US>

Fontpair

MDN Web Docs: CSS Box Model

<https://www.fontpair.co/?hl=en-US>

https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/The_box_model

CSS-Tricks: Guide to Flexbox

<https://css-tricks.com/snippets/css/a-guide-to-flexbox/>

CSS-Tricks: Guide to Grid

<https://css-tricks.com/snippets/css/complete-guide-grid/>